# Paper 13: Deterministic Time-Indexed Frame Synthesis Engine

Pillar: Graph-Loop Creative Pipelines — Autonomous Research Ecosystem

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## Abstract

We hypothesize that Deterministic Time-Indexed Frame Synthesis Engine achieves Hash-identical runs on TypeScript / Canvas while preserving strict mathematical invariants across edge, creative graph-loop, and decentralized clearing layers. We present a formal mathematical model, a production implementation in `TypeScript / Canvas`, and an automated empirical verification harness targeting **Hash-identical runs**. All benchmarks and invariants are reproducibly verified via our continuous Research Graph-Loop pipeline [3, 6, 2].

## 1 Introduction & Formal Hypothesis

Modern autonomous agent architectures face fundamental bottlenecks across constrained edge silicon, deterministic multi-modal synthesis, and decentralized economic clearing. In this work (Paper 13 within the *Graph-Loop Creative Pipelines* pillar), we address these limitations directly.

**Core Hypothesis:** We hypothesize that Deterministic Time-Indexed Frame Synthesis Engine achieves Hash-identical runs on TypeScript / Canvas while preserving strict mathematical invariants across edge, creative graph-loop, and decentralized clearing layers.

## 2 Mathematical Formulation

We formulate our system objective and state transition invariant as:

$$\mathcal{L}_{\text{total}}(\theta) = \mathbb{E}_{x\sim\mathcal{D}}\left[\mathcal{D}_{\text{KL}}(P_{\text{target}} \parallel P_{\text{draft}})\right] + \lambda\|\Delta_{\text{state}}\|_2^2 \tag{1}$$

Subject to strict runtime verification bounds (`Hash-identical runs`) executed across our hybrid compute fabric (Local Edge Silicon, Homelab Silo Server, and FAU Research Computing HPC nodes) [4, 5, 1].

## 3 System Architecture & Implementation

The reference architecture is implemented in `TypeScript / Canvas` across the following work package modules:

- `packages/motion_engine/src/index.ts`
- `research_ecosystem/pillar2_creative.py`

## 4 Empirical Evaluation & Benchmarks

We executed the automated verification suite for Paper 13 and recorded the following empirical metrics (Table 1):

Table 1: Empirical Verification Results for Paper 13

| Metric / Invariant | Measured Value |
|---|---|
| hash_identical_runs | `True` |
| seam_position_error | `0.0` |
| seam_velocity_error | `0.0` |
| passed | `True` |

## 5 Conclusion

We have demonstrated that *Deterministic Time-Indexed Frame Synthesis Engine* satisfies its target specification (`Hash-identical runs`) with deterministic reproducibility.

# References


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